

Future Work: Sequential Localization with Prediction-Based Filtering

VPR Tutorial Project Notes

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Motivation

The current Visual Place Recognition (VPR) workflow is mainly frame-by-frame: for each incoming image, the system finds the best matching place in the map. This works as a baseline, but it can be unstable under viewpoint and perspective changes, motion blur, and repetitive-looking scenes.

A strong future-work direction is to move from *single-frame matching* to *sequential state estimation*. The key idea is to combine:

- what the robot **expects** to see next (from motion and map), and
- what the camera **actually** observes (from VPR inference).

This is exactly the setting where a Kalman-style filter is useful.

Core Idea from the Discussion

The professor's point can be summarized as:

“Do not localize each frame independently. Use robot motion and map geometry to predict the next pose and expected observation, then fuse that prediction with visual matching evidence over time.”

So instead of only asking:

- “Which map image best matches this frame?”

we ask:

- “Given where I was and how I moved, where should I be now?”
- “Does the camera observation agree with that prediction?”

What “Expectation” Means in Practice

At time t , assume the robot has pose state:

$$\mathbf{x}_t = [x_t, y_t, \theta_t, v_t, \omega_t]^\top$$

where (x, y) is 2D location, θ is heading, and v, ω are linear/angular velocity terms.

Using odometry or control input, a motion model predicts the next state:

$$\hat{\mathbf{x}}_{t+1}^- = f(\mathbf{x}_t, \mathbf{u}_t)$$

This is the **expectation**: where the robot likely is before seeing the next camera frame.

From this predicted pose and the 2D map, the system can also infer what place/region should be visible.

What “Observation” Means

At time $t + 1$, the camera frame is processed by VPR to produce evidence such as:

- best matching map index,
- similarity score,
- top- k candidate matches.

This acts as a noisy measurement $\mathbf{z}_{t+1}$ of location.

Why Kalman-Style Filtering Helps

A Kalman filter framework alternates two steps:

1. **Prediction**: propagate state using robot motion model.
2. **Update**: correct predicted state using visual measurement.

It also propagates uncertainty, so the system can decide how much to trust motion vs vision at each step.

Benefits for this project:

- **Temporal consistency**: fewer sudden pose jumps between unrelated map places.
- **Robustness to bad frames**: one poor visual match has limited impact.
- **Better navigation behavior**: smoother and physically plausible trajectories.
- **Perspective-change resilience**: sequence context compensates for viewpoint variation.

Conceptual Model for Future Implementation

A practical formulation would use:

- **Process model** (motion): predicts (x, y, θ) over time,
- **Measurement model** (vision): maps VPR match evidence to pose observation,
- **Fusion layer** (filter): combines both using uncertainty.

At each time step:

$$\begin{aligned}\hat{\mathbf{x}}_t^- &= f(\hat{\mathbf{x}}_{t-1}, \mathbf{u}_{t-1}) \\ \hat{\mathbf{x}}_t &= \hat{\mathbf{x}}_t^- + \mathbf{K}_t(\mathbf{z}_t - h(\hat{\mathbf{x}}_t^-))\end{aligned}$$

where $\mathbf{K}_t$ is the Kalman gain and $h(\cdot)$ maps state to expected measurement.

Important Note on Filter Choice

A standard linear Kalman filter assumes linear dynamics and Gaussian noise. In real VPR localization:

- pose and camera observation relationships are often nonlinear,
- match hypotheses may be multi-modal in ambiguous scenes.

So likely candidates are:

- **EKF** (Extended Kalman Filter) as a strong baseline,
- **UKF** (Unscented Kalman Filter) for stronger nonlinear behavior,
- **Particle filter** if ambiguity is highly multi-modal.

How to Position This as Future Work

A concise statement for report/presentation:

“Future work will integrate VPR with probabilistic sequential localization in a 2D mapped environment. Motion/odometry will provide state prediction, while VPR similarity will provide measurement updates. We will evaluate gains in trajectory stability, relocalization accuracy, and robustness under viewpoint and perspective changes.”

Expected Evaluation Criteria

When this is implemented, useful evaluation metrics include:

- localization error over trajectory (position and heading),
- temporal smoothness / trajectory consistency,
- relocalization robustness after visual degradation,
- precision-recall style place-recognition metrics before and after filtering.